SUMMARY TABLE: PHYSICS GRADE 10

Sub-Strand 1.4: Energy, Work, Power and Machines — Teacher Reference (Pre-filled)

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Sub-Strand	Sub-Strand 1.4: Energy, Work, Power and Machines
Driving Question	DRIVING QUESTION: Why can a jua kali mechanic in Nairobi loosen a wheel nut effortlessly with a long spanner when a short one fails — and where does the energy go when friction makes the nut hot?

INSTRUCTIONS
FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 **Kinetic Energy and Gravitational Potential Energy: What Makes Things Move and Fall?**	Students observed a ball rolled across a desk and a heavy textbook lifted to different heights. They noted that a faster-moving ball caused a greater dent in soft clay (evidence of more energy of motion), and that a book dropped from a greater height made a louder impact (evidence of more stored energy due to position). In a Kenya-context demonstration, a maize cob dropped from shoulder height versus knee height showed a measurable difference in impact. Teachers should prompt students to record both qualitative observations and any measurements taken (mass, height, speed estimate).	Learners discovered two distinct forms of mechanical energy: (1) Kinetic Energy — the energy an object possesses because it is moving, which increases with both mass and speed; and (2) Gravitational Potential Energy — the energy stored in an object because of its height above a reference point, which increases with both mass and height. They recognised that energy is not visible but its effects are measurable, laying the conceptual groundwork for all subsequent energy lessons.	$KE = \frac{1}{2}mv^2$ — kinetic energy (in Joules) equals half the product of mass (kg) and the square of velocity (m/s). $PE = mgh$ — gravitational potential energy equals mass × acceleration due to gravity (10 m/s^2 in Kenya-standard problems) × height (m). Pedagogical note: Emphasise that v is squared, so doubling speed quadruples KE — a common misconception to address early. Connect to the driving question: the mechanic's swinging arm and the heavy spanner both possess KE; the raised spanner possesses PE. Both forms will matter when work is introduced in Lesson 3.
Lesson 2 **Conservation of Mechanical Energy: The Invisible Transfer**	Students observed a pendulum (a stone tied to a string, as a locally available model) swung from a marked height. They tracked the stone's speed at the bottom (fastest) and its height at the opposite side (almost equal to the release height). A second observation used a video or teacher demonstration of a cyclist freewheeling down a slope in Nairobi's Ngong Hills — fast at the bottom, slow near the top.	Learners discovered the Law of Conservation of Mechanical Energy: in the absence of friction and air resistance, the total mechanical energy (KE + PE) of a system remains constant. Energy is not created or destroyed — it is transferred between kinetic and potential forms. When the stone swings to its lowest point, all PE has converted to KE; at the highest point, all KE has converted back to PE. This is an	Total Mechanical Energy $E = KE + PE = \frac{1}{2}mv^2 + mgh$ = constant (frictionless system). At any two points: $\frac{1}{2}mv_1^2 + mgh_1 = \frac{1}{2}mv_2^2 + mgh_2$. Pedagogical note: Explicitly flag the 'frictionless' assumption and ask students to predict what happens in the real world (energy appears to be 'lost' — this sets up Lessons 3 and 7). Connect to the driving question: the mechanic does work against friction; the nut gets hot because

	Students recorded height and estimated speed at multiple points, noting the trade-off between the two quantities.	idealised (frictionless) model — a critical limitation students must carry forward.	mechanical energy is being converted to thermal energy — conservation still holds, but the energy leaves the mechanical system. Teachers should cold-call at least 3 students to articulate where energy 'goes' before moving on.
Lesson 3 **Work: Definition and Calculation**	Students observed two scenarios: (1) a learner pushing a desk across the classroom floor (force in the direction of motion) versus (2) a learner carrying a heavy bag of githeri on their head while walking horizontally (force upward, motion horizontal). They used a spring balance to measure the pushing force and a metre rule to measure displacement. They compared the two scenarios and discussed whether 'effort' always equals 'work done' in the physics sense.	Learners discovered that in physics, work is only done when a force causes displacement in the direction of the force. Carrying a load horizontally does zero work against gravity because the force (upward) is perpendicular to the displacement (horizontal). Work depends on three factors: the magnitude of the force, the magnitude of the displacement, and the angle between them. This precise definition often contradicts everyday use of the word 'work,' which is a key conceptual hurdle.	$W = Fd\cos\theta$, where W = work done (Joules, J), F = applied force (Newtons, N), d = displacement (metres, m), and θ = angle between the force and displacement vectors. When $\theta = 0°$, $\cos\theta = 1$, so $W = Fd$ (maximum positive work). When $\theta = 90°$, $\cos\theta = 0$, so $W = 0$ (no work done). When $\theta = 180°$, $\cos\theta = -1$, so $W = -Fd$ (negative work, e.g. friction opposing motion). Pedagogical note: Use the jua kali spanner directly here — the mechanic applies force along the arc of the spanner's swing; displacement is along that arc; $\theta \approx 0°$ for maximum work on the nut. This is the first direct bridge to the driving question. Ensure students practise at least three calculation examples with varying angles.
Lesson 4 **The Work–Energy Theorem: Connecting Work to Changes in Energy**	Students observed a toy car pushed across a smooth surface with a measured force over a measured distance, then checked its speed using a timer or motion sensor. They compared the work done on the car ($W = Fd$) with the change in its kinetic energy ($\Delta KE = \frac{1}{2}mv^2 - \frac{1}{2}mu^2$). A second scenario showed a car braking to a stop — friction did negative work, reducing KE to zero. Teachers should record both the calculated W and measured ΔKE on the board for direct comparison, highlighting any small discrepancies due to real-world friction.	Learners discovered the Work–Energy Theorem: the net work done on an object equals the change in its kinetic energy. This theorem unifies the concepts from Lessons 1–3. Positive net work increases KE (object speeds up); negative net work decreases KE (object slows down). When a mechanic applies force through a spanner, work is done and energy is transferred — part of it goes into rotating the nut (useful work), part into heating the nut and spanner (work done against friction).	$W_{net} = \Delta KE = \frac{1}{2}mv^2 - \frac{1}{2}mu^2$, where u = initial speed and v = final speed. This can be rewritten as: $W_{useful} + W_{friction} = \Delta KE$. In a real system, $W_{friction}$ is always negative (friction removes energy from the mechanical system and converts it to thermal energy). Pedagogical note: This lesson is the conceptual pivot of the sub-strand. Ask students to revise their Driving Question Board entry: 'Where does the energy go when friction makes the nut hot?' — they should now be able to say that friction does negative work on the nut, converting mechanical energy to thermal (internal) energy. Cold-call at least 4 students to explain this in their own words before proceeding.
Lesson 5 **Power: Definition, Calculation, and Everyday Applications**	Students timed themselves climbing a staircase (or stepping up a measured height on a bench) and recorded their weight, height climbed, and time taken. Different students completed the same climb in	Learners discovered that power is not about how much work is done, but about how quickly work is done. Two people can do identical amounts of work (climb the same height) yet have very different power	$P = W/t$, where P = power (Watts, W), W = work done (Joules), t = time (seconds). Since $W = Fd$, this can also be written as $P = Fv$, where v = velocity — useful for problems involving constant force and

	different times. They also compared a jiko (charcoal stove) boiling water slowly versus an electric kettle (where available) boiling water quickly — same task, different rates. Teachers should ensure each student calculates their own weight (mg) and records time carefully for subsequent calculation.	outputs if they take different amounts of time. A more powerful machine or person completes the same task faster. Power is measured in Watts (W), where 1 Watt = 1 Joule per second. This concept reframes the driving question: a mechanic using a long spanner may not do more total work, but the mechanical advantage changes how the effort force is applied.	speed (e.g., a matatu engine maintaining speed on a Nairobi highway). Pedagogical note: Distinguish clearly between 'power' (rate of work) and 'energy' (total work done) — students frequently conflate these. Use the staircase data: a student who climbed faster had greater power but did the same work as a slower student. Assign at least two calculation problems involving Kenyan contexts (e.g., a porter at Mombasa port, a Rift Valley farmer carrying produce).
Lesson 6 **Simple Machines: Types and Mechanical Advantage**	Students examined and manipulated locally available simple machines: a metal rod used as a lever to lift a heavy stone (class 1 lever), a ramp made from a plank for lifting a jerrycan, a wheel and axle modelled with a spool, a pulley rigged with rope, a metal wedge (like a panga blade), and a bolt and nut as a screw. For each machine, they measured the load lifted and the effort applied using a spring balance. They recorded both values and noted that in every case where the load exceeded the effort, the effort moved through a greater distance.	Learners discovered the six types of simple machines (lever, inclined plane, pulley, wheel and axle, wedge, screw) and the concept of Mechanical Advantage (MA). MA > 1 means the machine multiplies force — a smaller effort lifts a larger load. However, there is a trade-off: the effort always moves through a greater distance than the load. The jua kali spanner is a lever (wheel and axle / lever class); a longer arm gives greater MA, which is precisely why the mechanic can loosen the nut with less effort.	Mechanical Advantage (MA) = Load (L) ÷ Effort (E). For the spanner as a lever: MA = effort arm length ÷ load arm length (or equivalently, the longer the spanner, the greater the MA). Pedagogical note: This is the direct answer to the first half of the driving question. Teachers should explicitly state: 'The long spanner gives a higher MA — the mechanic applies a small force over a long distance to produce a large turning force (torque) on the nut.' Cold-call 3 students to calculate MA for a spanner of 30 cm vs. 10 cm if the nut requires 120 N to turn (expected answers: MA = 4 and MA = 1.33 respectively). Emphasise that MA is dimensionless (no units).
Lesson 7 **Velocity Ratio, Efficiency, and the Cost of Friction**	Students used a pulley system or an inclined plane to lift a load, carefully measuring both the effort distance (how far the rope was pulled / how far the effort moved) and the load distance (how far the load moved). They then compared the theoretical MA (from Lesson 6) with the actual MA measured in the experiment. In every case, the actual MA was lower than the theoretical value. Students also felt the axle of the pulley or the surface of the ramp after repeated use — it was noticeably warm. Teachers should draw explicit attention to this warmth as physical evidence of energy conversion.	Learners discovered Velocity Ratio (VR) — the ratio of the distance moved by the effort to the distance moved by the load — and that VR is determined purely by the geometry of the machine (it does not change with friction). Efficiency measures how much of the input work becomes useful output work. Because friction always converts some mechanical energy to heat, no real machine achieves 100% efficiency. The warm pulley axle and warm nut are direct evidence of this energy conversion, fully answering the second half of the driving question.	VR = effort distance ÷ load distance (dimensionless). Efficiency (%) = (MA ÷ VR) × 100% = (Work output ÷ Work input) × 100%. Because friction always makes MA < VR in real machines, efficiency is always less than 100%. The 'lost' energy is not destroyed — it is converted to thermal energy (the nut heats up). Pedagogical note: This lesson resolves the second part of the driving question entirely. Write on the board: 'Energy input = Useful work output + Energy lost to friction (heat).' Ask students to update their cumulative model diagram to show the energy pathway. Ensure students can substitute values and calculate efficiency for at least two machine types.
Lesson 8 **Synthesis, Model**	Students revisited the original anchoring phenomenon: a jua kali mechanic in Nairobi	Learners synthesised the complete explanation: (1) The long spanner acts as a	The integrated explanation links every concept: KE and PE (Lesson 1) →

Finalisation, and DQB Resolution: The Full Story of the Jua Kali Mechanic	using a long spanner versus a short one. They reviewed all evidence collected across Lessons 1–7: energy forms (KE, PE), energy conservation, the definition of work, the Work–Energy Theorem, power, mechanical advantage, velocity ratio, and efficiency. Using their cumulative model diagrams and Driving Question Board, they identified which concepts answered which part of the driving question. Teachers should display all DQB questions and systematically check off each one as students explain the resolution.	lever with a large effort arm, giving a high Mechanical Advantage — a small effort force produces a large torque on the nut. (2) The mechanic does work on the nut ($W = Fd \cos\theta$). (3) Because the nut-thread system has friction, not all work input becomes useful rotational work — some is converted to thermal energy, heating the nut. (4) Efficiency < 100% in all real machines due to friction. (5) Energy is conserved throughout: mechanical energy input = useful work on nut + heat generated by friction. The full energy story is complete.	Conservation of Energy (Lesson 2) → Work = $Fd \cos\theta$ (Lesson 3) → Work–Energy Theorem, friction converts energy to heat (Lesson 4) → Power = W/t (Lesson 5) → MA = Load/Effort, long spanner gives high MA (Lesson 6) → VR, Efficiency < 100%, friction accounts for missing energy (Lesson 7) → Full synthesis (Lesson 8). Pedagogical note: This is the assessment and celebration lesson. Teachers should cold-call at minimum 5 different students to each explain one concept link in the chain. Students should submit or present their finalised model diagrams. A strong final model will show: effort force → spanner (lever) → torque on nut → nut rotation (useful work output) + heat at nut-thread contact (friction loss), with energy values labelled and efficiency calculated. Affirm that the jua kali mechanic intuitively applies physics that students can now explain quantitatively.